Computer Networks LAB-01

Anurag Kashyap

Experiment Title

→ Study of LAN Topology, Network devices, and Address configuration.

Objective:

To study LAN Topology, identify networking devices, analyze wired and wireless communication measure network speed and obtain Network.

Problem Statement

1) Draw the Network Topology.

- Identify and draw the LAN Topology of our college (eg: Star, Bus, Mess)
- clearly label all components, such as Router, Switch, Access point and end devices (PCs, laptop, etc)

Soln: The college campus Network examined during this study includes the following components.

-> layer-3 core switches installed in the server room for Network Management and routing.

-> POE (Power over Ethernet) switches used to supply power and connectivity to network devices.

-> wireless Access point (APs) for providing wifi connectivity Across the campus.

=> Ethernet ports available in classrooms for wired Network Access

The college campus Networks uses a Hybrid Hierarchical Topology that contains / combines:

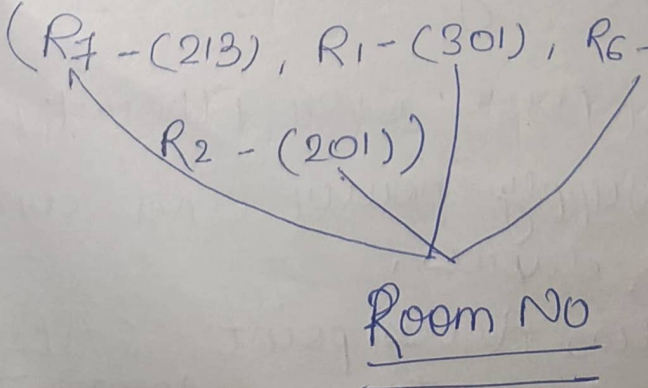
- Tree Topology
- Extended star Topology
- Bus Topology
- Ring Topology.

The Network is divided into three layers

1) core layer
2) Distribution layer
3) Access layer

The core layer contains networking devices such as firewall router, layer-3 switches, patch rack and servers

And The distribution layer consists of four switches placed in rooms/ floors of the main building for Network distribution



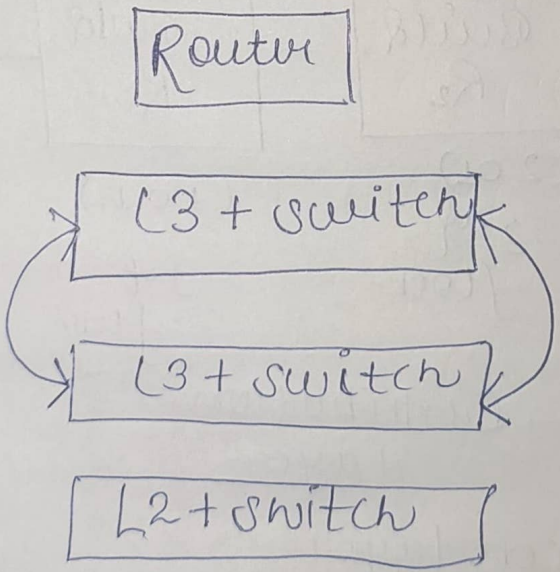
The Access layer contain Three ~~layer~~ switches located in different rooms of the main building

- R_1 - Room No - 301 (3rd floor)
- R_4 - Room No - 05 (Ground floor)
- R_5 - Room No - 18 (Ground floor)

Access layer directly connect.

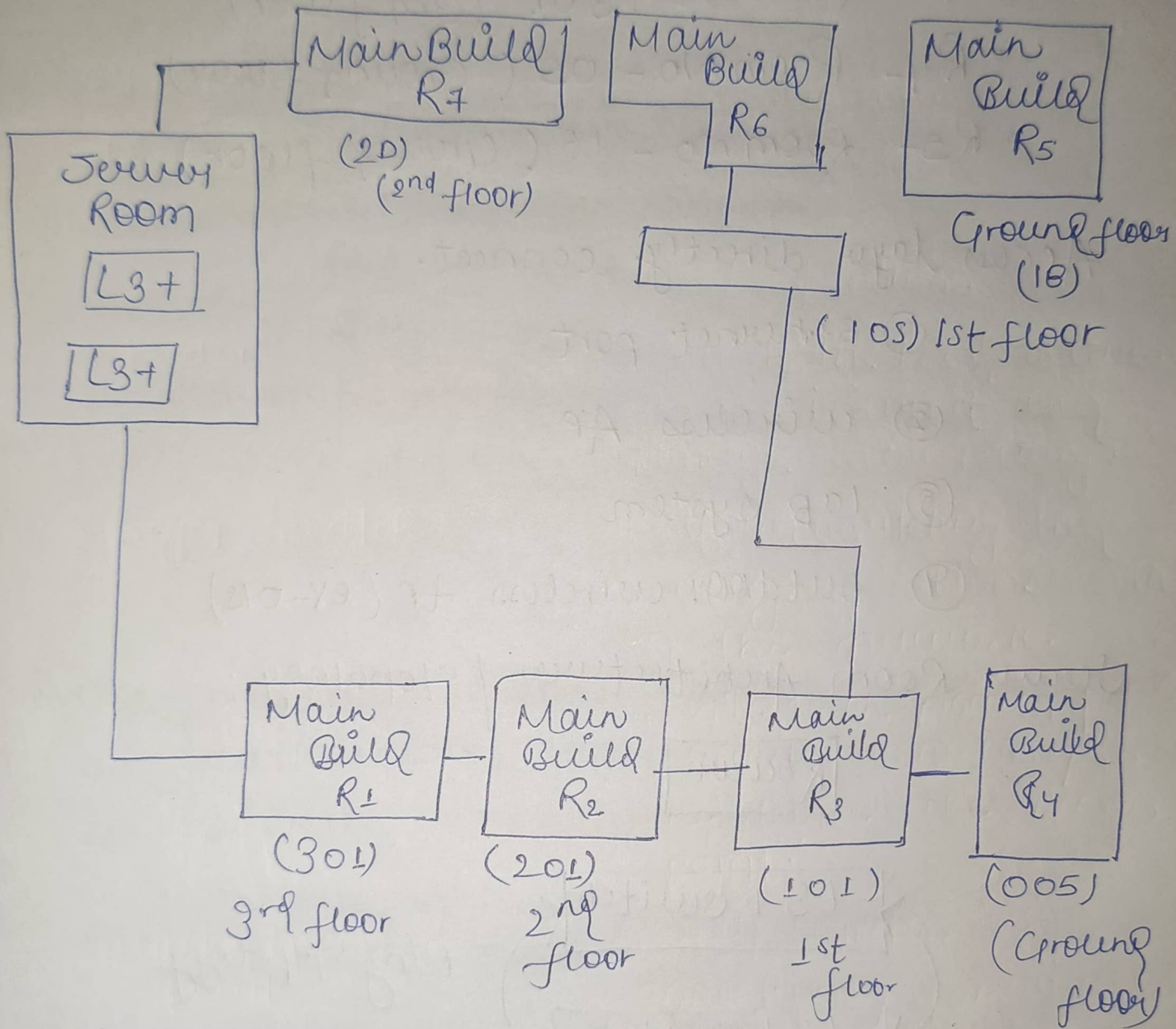
1. Ethernet port
2. wireless AP
3. LAB system
4. outdoor wireless AP (ex-01s)

Server Room Architecture / Topology:



Ring Topology

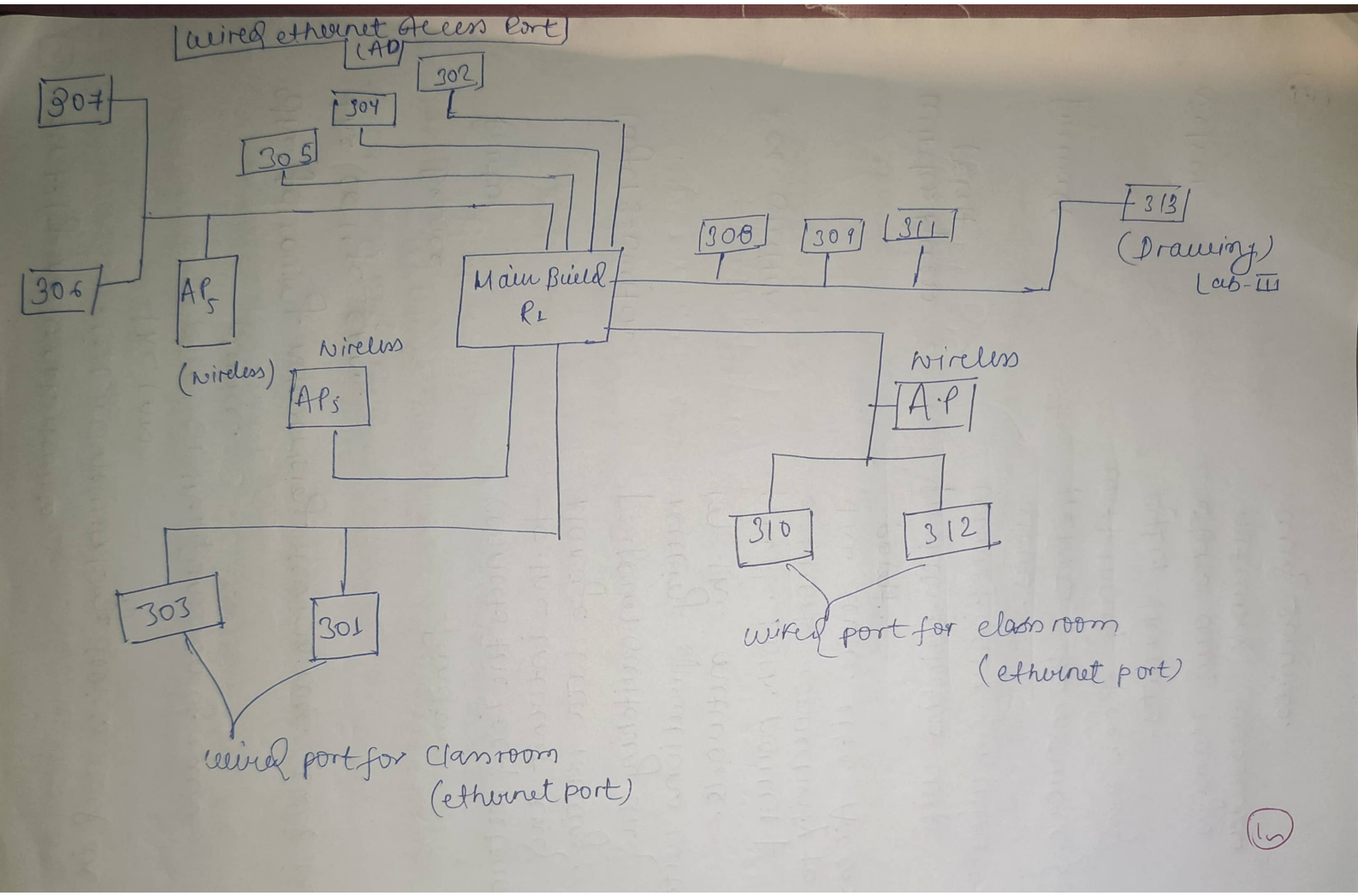
Topology



→ R1, R7, R2, R6 — Distribution layer

→ Server Room — core layer

→ R1, R2, R3, R4, R5, R6, R7 — Access point.


wired ethernet Access Port
LAD
302
304
305
307
306
APs
(Wireless)
Wireless
APs
Main Build
RL
308
309
311
313
(Drawing)
Lab-III
Wireless
A.P
310
312
303
301
wired port for Classroom
(ethernet port)
wired port for classroom
(ethernet port)

② Identify Network devices

- list all networking devices used in the LAN
- Mention the function of each device.

The following networking devices are used in the college CAN.

Device	function.
Router	connects the local network to the internet and Manage data routing.
layer 3-switch	perform switching and routing operations with in the network.
POE switch	provide both power & network connectivity to the device like Access point.
wireless Access point	provides ~~wireless~~ wifi connectivity for wireless users.
firewall	Protects the network from unAuthored Access and cyber threats.
Server	stores and Manages network resources and services.

Ethernet cable	used for wired communication between devices.
end Devices (PCs/ laptop)	used by users to a access network resources and internet services.

③ Analyse connection Type

The college campus networks use both wired (ethernet) and wireless (wifi) connections for communications.

* wired connections are used in:
  * *) computer laboratories.
  * *) Server room
  * → Desktop Computers
  * → Network switches & routers

→ ethernet cable are used to provide stable and high speed communications.

→ wireless connections (wifi) are used in:

- ◦) Laptop
- ◦) Mobile phone
- ◦) Smart phone
- ◦) Classroom & campus Areas.

```
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ANURAG KUMAR SINGH>ipconfig

Windows IP Configuration


Unknown adapter Local Area Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 11:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:40e4:4c:c507:fab7:3520:99ee:a91
   Temporary IPv6 Address. . . . . . : 2409:40e4:4c:c507:2937:b16b:4fad:fbc0
   Link-local IPv6 Address . . . . . : fe80::f35b:e900:a275:b7b9%19
   IPv4 Address. . . . . . . . . . . : 10.52.42.65
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::7c4f:d9ff:fec4:ebac%19
                                       10.52.42.177

C:\Users\ANURAG KUMAR SINGH>
```

```
C:\Users\ANURAG KUMAR SINGH>getmac

Physical Address    Transport Name
=================== ==========================================================
E0-C2-64-9E-94-09   \Device\Tcpip_{EFB4184E-BC1E-4DC0-8B8E-DE6A073ED12F}
00-FF-0D-E0-4C-3C   Media disconnected

C:\Users\ANURAG KUMAR SINGH>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : LAPTOP-JB0Q8EER
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Unknown adapter Local Area Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : TAP-Windows Adapter V9
   Physical Address. . . . . . . . . : 00-FF-0D-E0-4C-3C
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : E0-C2-64-9E-94-0A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 11:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #3
   Physical Address. . . . . . . . . : E2-C2-64-9E-94-09
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
   Physical Address. . . . . . . . . : E0-C2-64-9E-94-09
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2409:40e4:4c:c507:fab7:3520:99ee:a91(Preferred)
   Temporary IPv6 Address. . . . . . : 2409:40e4:4c:c507:2937:b16b:4fad:fbc0(Preferred)
   Link-local IPv6 Address . . . . . : fe80::f35b:e900:a275:b7b9%19(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.52.42.65(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 31 May 2026 16:52:00
   Lease Expires . . . . . . . . . . : 31 May 2026 17:51:59
   Default Gateway . . . . . . . . . : fe80::7c4f:d9ff:fec4:ebac%19
                                       10.52.42.177
   DHCP Server . . . . . . . . . . . : 10.52.42.177
   DHCPv6 IAID . . . . . . . . . . . : 216056420
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2E-B8-3D-FE-E0-C2-64-9E-94-09
   DNS Servers . . . . . . . . . . . : 10.52.42.177
                                       2409:40e4:4c:c507::4b
   NetBIOS over Tcpip. . . . . . . . : Enabled

C:\Users\ANURAG KUMAR SINGH>
```

```
C:\Users\ANURAG KUMAR SINGH>netsh wlan show interfaces

There is 1 interface on the system:

    Name                   : Wi-Fi
    Description            : Intel(R) Wi-Fi 6E AX211 160MHz
    GUID                   : efb4184e-bc1e-4dc0-8b8e-de6a073ed12f
    Physical address       : e0:c2:64:9e:94:09
    Interface type         : Primary
    State                  : connected
    SSID                   : Anurag
    AP BSSID               : 7e:4f:d9:c4:eb:ac
    Band                   : 2.4 GHz
    Channel                : 7
    Connected Akm-cipher   : [ akm = 00-0f-ac:08, cipher =  00-0f-ac:04 ]
    Network type           : Infrastructure
    Radio type             : 802.11n
    Authentication         : WPA3-Personal
    Cipher                 : CCMP
    Connection mode        : Profile
    Receive rate (Mbps)    : 78
    Transmit rate (Mbps)   : 72.2
    Signal                 : 96%
    Rssi                   : -34
    Profile                : Anurag
    QoS MSCS Configured          : 0
    QoS Map Configured           : 0
    QoS Map Allowed by Policy    : 0


C:\Users\ANURAG KUMAR SINGH>
```

→ Wifi Access Point (APs) Provide wireless internet connectively accross the campus.

(4) Measure Network speed.

- find the link speed of
  - → wired network (if available)
  - → wireless network
- Mention the tools / commands used

Q: Task Manager, show interface etc.

Ans: link speed is shown in the images in the next page at both wired and wireless.

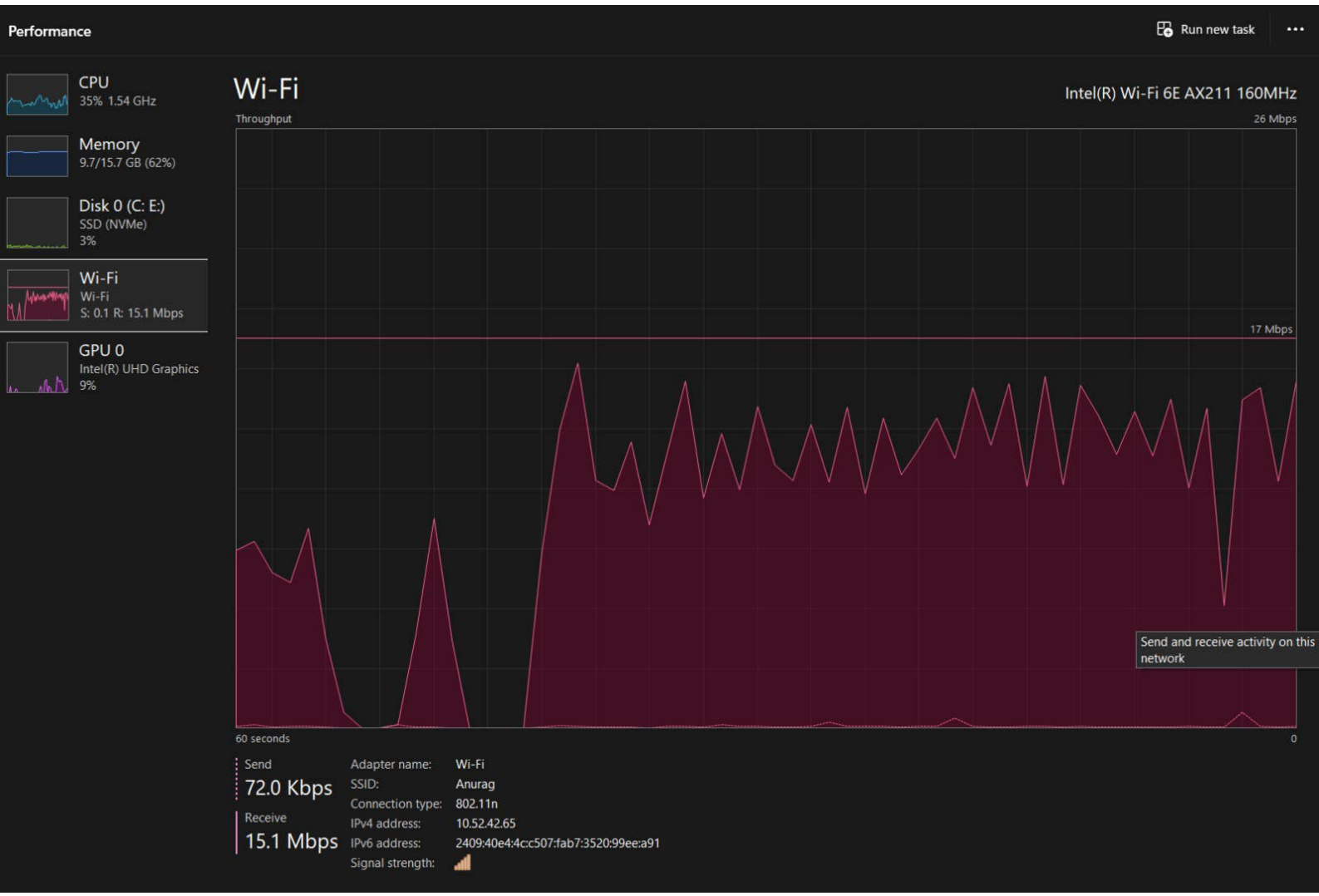
The tools/ commands used to check the speed using:

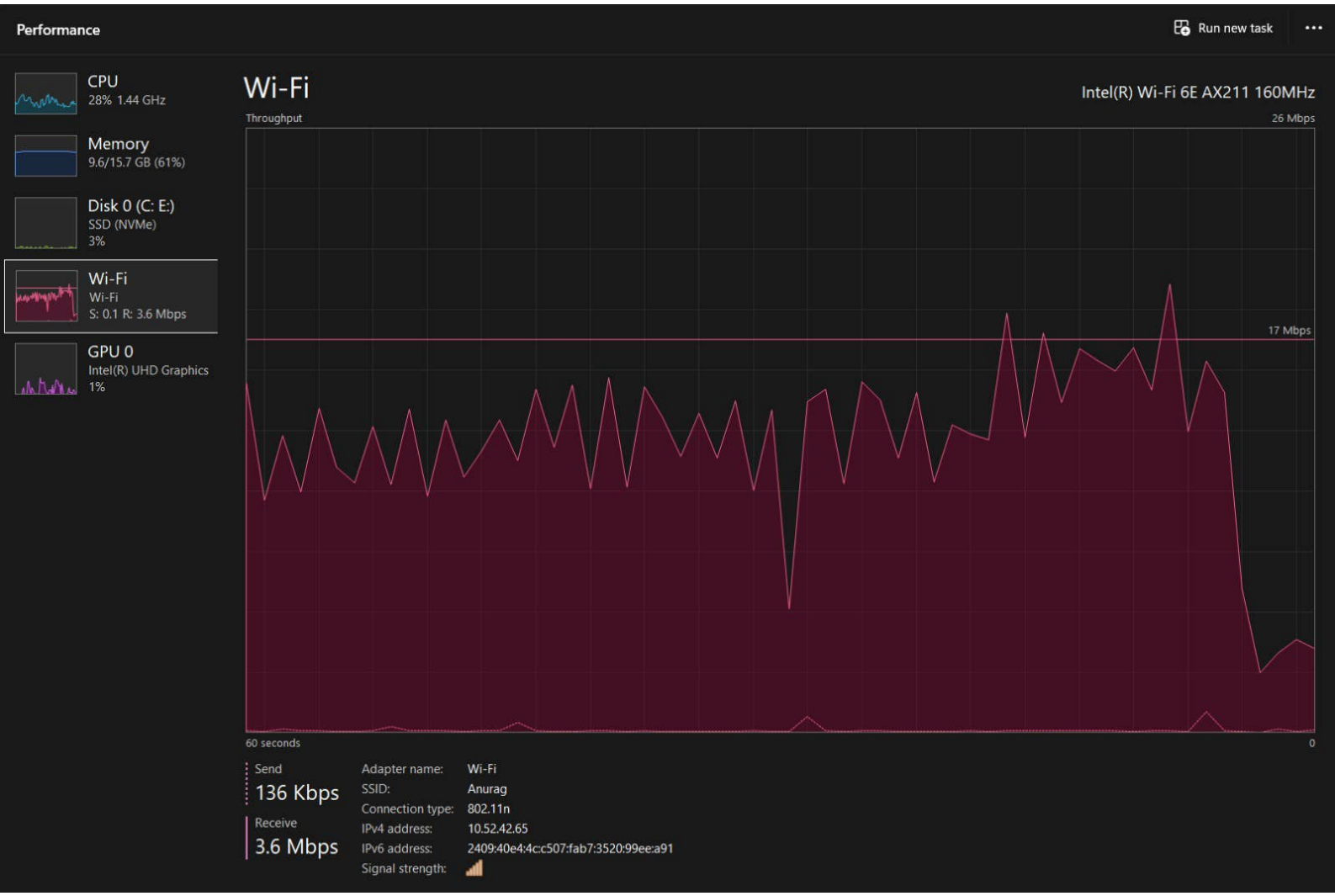
-) Task Manager
-) command prompt

Command used:

cmd

netsh wlan show interfaces.


Performance
Run new task
CPU
35% 1.54 GHz
Memory
9.7/15.7 GB (62%)
Disk 0 (C: E:)
SSD (NVMe)
3%
Wi-Fi
Wi-Fi
S: 0.1 R: 15.1 Mbps
GPU 0
Intel(R) UHD Graphics
9%
Wi-Fi
Intel(R) Wi-Fi 6E AX211 160MHz
Throughput
26 Mbps
17 Mbps
Send and receive activity on this network
60 seconds
0
Send
72.0 Kbps
Receive
15.1 Mbps
Adapter name: Wi-Fi
SSID: Anurag
Connection type: 802.11n
IPv4 address: 10.52.42.65
IPv6 address: 2409:40e4:4cc507:fab7:3520:99ee:a91
Signal strength:


Performance
Run new task
CPU
28% 1.44 GHz
Memory
9.6/15.7 GB (61%)
Disk 0 (C: E:)
SSD (NVMe)
3%
Wi-Fi
Wi-Fi
S: 0.1 R: 3.6 Mbps
GPU 0
Intel(R) UHD Graphics
1%
Wi-Fi
Intel(R) Wi-Fi 6E AX211 160MHz
Throughput
26 Mbps
17 Mbps
60 seconds
0
Send
136 Kbps
Receive
3.6 Mbps
Adapter name: Wi-Fi
SSID: Anurag
Connection type: 802.11n
IPv4 address: 10.52.42.65
IPv6 address: 2409:40e4:4cc507:fab7:3520:99ee:a91
Signal strength:

⑤ find system network details

+ Determine and reward the following of your computer

- MAC address
- I/P Address (IPV4)
- Subnet Mask.

• Mention the command (4) used

Ans:- The reward like Mac Address, I/P Address (IPV4) and subnet Mask shown in the next page.

The command is used :

+ To find I/P Address and Subnet Mask.

→ ip config

+ To find MAC Address

→ get mac

⑥ Difference between wired and wireless speed:-

These are various diff between wired and wireless speed:

Wired Network

→ Uses ethernet cable for communication

→ Generally provides higher speed

→ More stable & reliable

→ Lower latency

→ Better for gaming and large file transfer

→ Less affected by distance

Wireless Network

→ Uses wifi radio signals

→ Speed may fluctuate

→ Affected by signal interference.

→ Higher latency

→ Better for mobility and convenience

→ Speed decreases with distance.

▷ Role of Subnet Mask in networking.

A subnet mask is used to divide an IP address into two parts:-

* network portion
* Host "

It helps devices identify whether another device is on the same network or a diff network.

function of Subnet Mask :-

* identifies the network and host address
* controls network traffic
* Help in proper routing of data
* Improves network management and security

ex :- IPv4 Address : [redacted] 10.52.42.65

Subnet Mask : 255. 255. 255. 0

Conclusion :-

The experiment helped us in understanding LAN Topology networking devices and network configuration parameters, different connection types such as wired and wireless networks were studied, commands were used for identify IP address, MAC address, Subnet Mask and network speed

This experiment provides practical knowledge of computer fundamentals.